

Data Governance, Data Privacy, and Blockchain / Web3 Trends

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Course Map

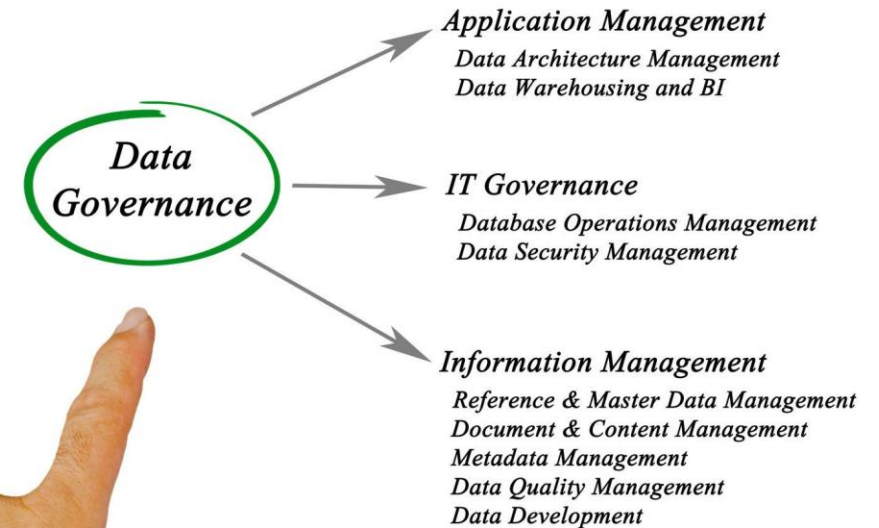
Week Number	Inclusive Dates	Module Number	Assessments	Topics
1	14 Feb	1	-	Course Introduction, Discussion of Deliverables, Framework for Analyzing IT Trends, Global IT Landscape
2-3	21 Feb, 28 Feb	2	Analysis Paper (I)	Artificial Intelligence (AI) & Generative AI, Large Language Models, Enterprise AI Adoption, AI Ethics, Bias and Explainability, AI Regulation (EU AI Act, PH AI initiatives)
4-5	7 Mar, 14 Mar	3	Presentation (G)	Cybersecurity Trends, Zero Trust, and Cloud-Native Security
6-7	21 Mar, 28 Mar	4	Review Paper (I)	Data Governance, Data Privacy, and Blockchain / Web3 Trends
8	4 Apr	5	Plan Report (G)	Internet of Things (IoT), Smart Systems & Industry 4.0, IoT Security Challenges
9	11 Apr	-	Exam	MIDTERM EXAMINATION
10-11	18 Apr 25 Apr	6	Research Paper (I)	Automation & Hyperautomation, RPA & Process Mining, Low-Code / No-Code Platforms, Quantum Computing, Post-Quantum Cryptography
12-13	2 May 9 May	7	Proposal Paper (G)	Sustainable IT, Green Computing, ESG & Digital Responsibility, DevOps & DevSecOps, Platform Engineering, GitOps
14-15	16 May 23 May	8	Analysis Report (I)	Extended Reality (AR/VR/MR), Spatial Computing, Human-Computer Interaction Trends, Digital Transformation in the Philippines, eGovernment, FinTech & Digital Banking
16-17	30 May 6 Jun	9	Presentation (G)	IT Project Failures, Governance Breakdowns, Ethical and Strategic Risks
18	13 Jun	-	Exam	FINALEXAMINATION



Learning Objectives

In the end of this session, students will be able to:

- Explain the concept of data governance
- Identify principles of data privacy and protection
- Describe major global and Philippine data regulations
- Understand blockchain and decentralized technologies
- Evaluate the opportunities and risks of Web3 systems

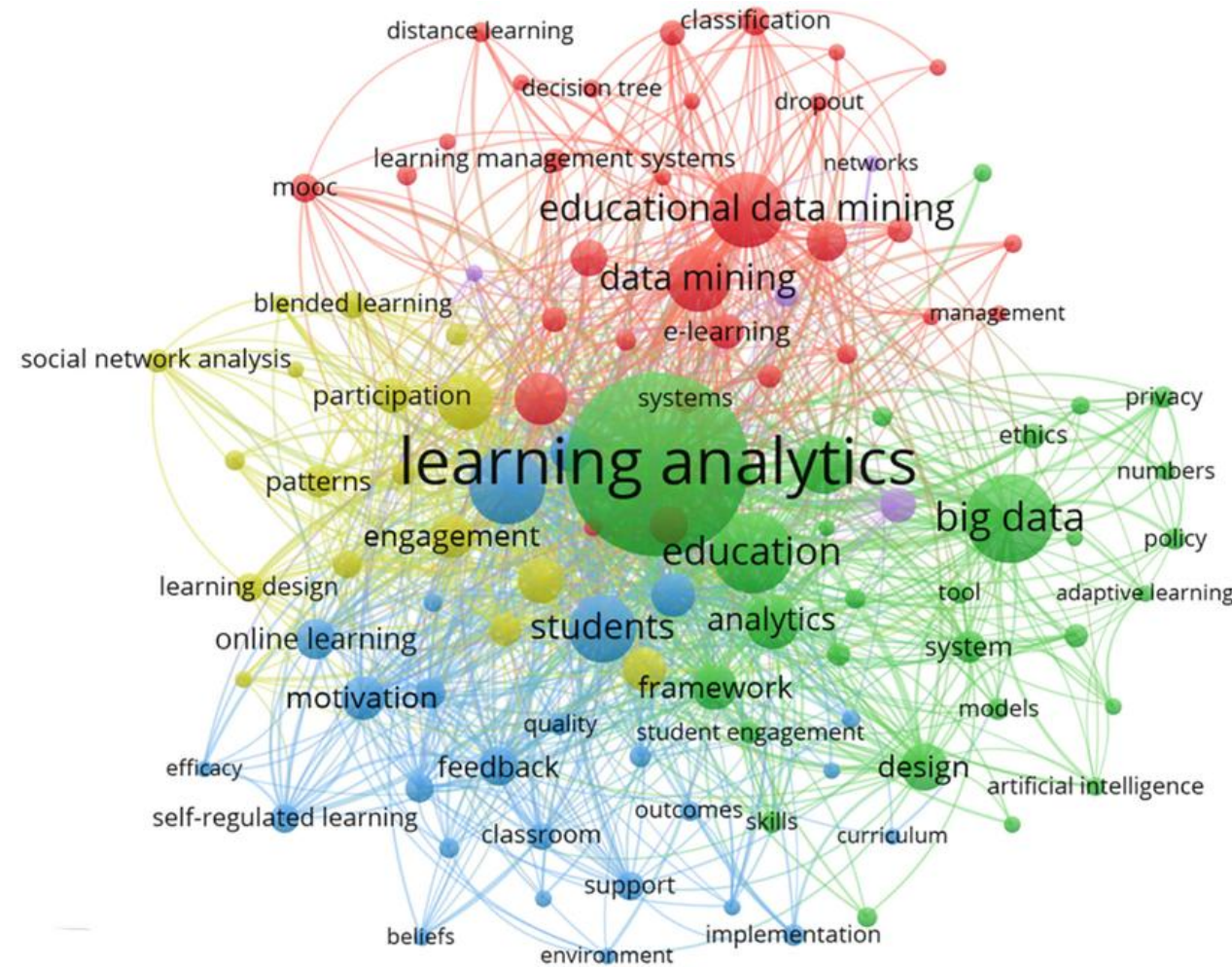


What is Data?

Every digital interaction generates data. Organizations use this data to improve decisions and services.

Data:

- Raw facts or information
- Can be structured or unstructured
- Used for analysis and decision making

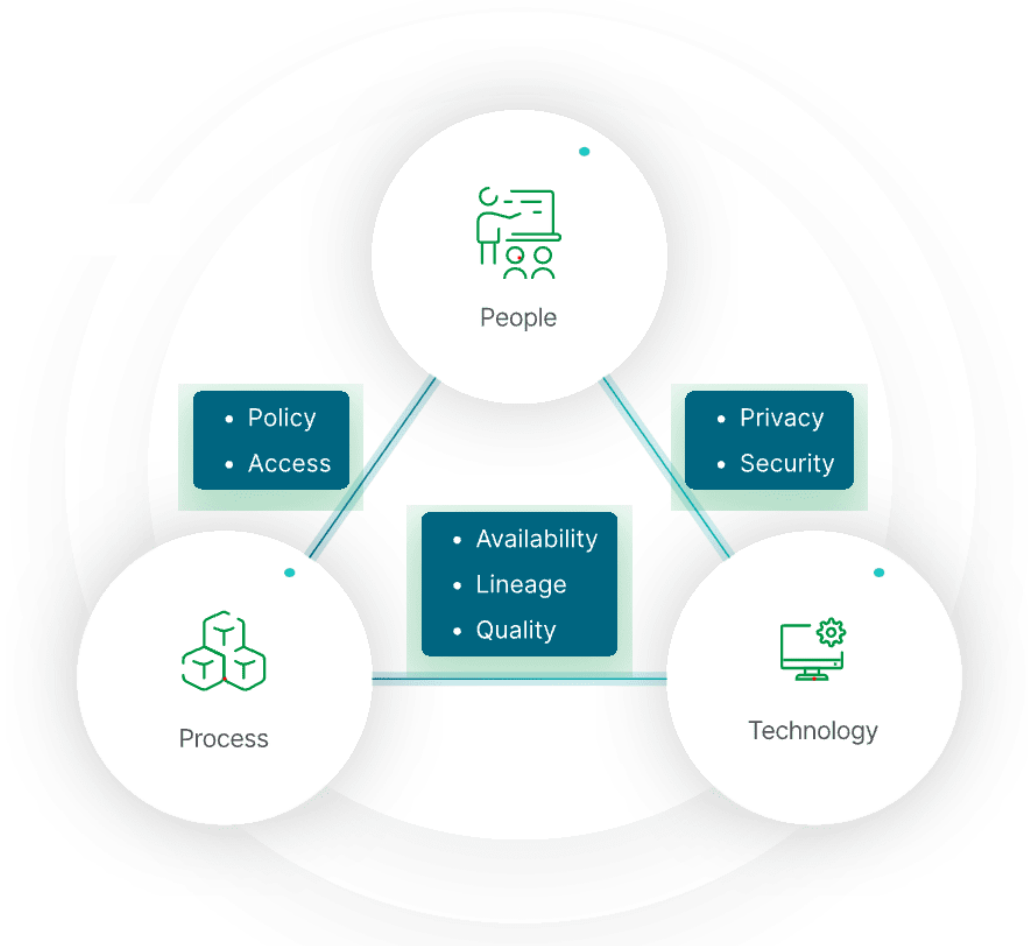


What is Data Governance?

Organizations must manage data responsibly and systematically.

Data Governance:

- Policies for managing data assets
- Ensures data quality and security
- Defines roles and responsibilities



Why Data Governance Matters?

Poor data governance can lead to security breaches, legal problems, and incorrect business decisions.

Importance:

- Protects sensitive data
- Improves data accuracy
- Supports regulatory compliance



Types of Organizational Data

Organizations handle different categories of data.

Data Types:

- Personal data – information about individuals
- Operational data – internal business information
- Financial data – monetary transactions
- Analytical data – insights derived from analysis

What is personal data?



The infographic features a central illustration of a person in a blue suit holding a tablet. To the left of the person is a vertical list of data categories, each with an icon and text: 'Name' (person icon), 'Address' (house icon), 'Localisation' (location pin icon), 'Online identifier' (person head icon), 'Health information' (heart with pulse icon), 'Income' (Euro symbol icon), 'Cultural profile' (clapping hands icon), and 'and more' (three dots icon). To the right of the person, the text 'COLLECT STORE USE DATA?' is displayed in large, bold, orange letters, followed by a downward-pointing blue arrow and the sentence 'You have to abide by the rules.' in white text.

Icon	Data Type
Person icon	Name
House icon	Address
Location pin icon	Localisation
Person head icon	Online identifier
Heart with pulse icon	Health information
Euro symbol icon	Income
Clapping hands icon	Cultural profile
Three dots icon	and more

**COLLECT
STORE
USE
DATA?**

▼

You have to abide by the rules.



Data Privacy

Data privacy focuses on protecting personal information.

Data Privacy:

- Protection of personal data
- Limits unauthorized access
- Ensures informed consent



Philippine Data Privacy Act

The Philippines has legal protections for personal data.

Data Privacy Act (RA 10173):

- Protects personal information
- Requires consent for data processing
- Establishes National Privacy Commission



POLYTECHNIC UNIVERSITY OF THE PHILIPPINES
UNISAN, QUEZON BRANCH

RA
10173

DATA PRIVACY ACT

A.K.A. The Data Privacy Act (DPA), or Republic Act No. 10173

HOW DO WE COMPLY?

All **personal data** must:

- ✓ be collected with the **customers' proper consent**
- ✓ be solely **used for its intended purposes**
- ✓ have **legitimate reasons for collection** that must be clear to both parties giving and receiving information
- ✓ **be protected and secured**, from collection to proper disposal, avoiding access from unauthorized parties

Companies should have:

- ✓ a **Data Protection Officer**
- ✓ **Routine assessments** to ensure quality data protection
- ✓ **Privacy knowledge programs and privacy and data policies** to regulate the handling of information
- ✓ A **proper procedure for breach notification** to its customers

WHAT WOULD NON-COMPLIANCE ENTAIL?

Improper/unauthorized processing, handling or disposal of personal information can be penalized by



IMPRISONMENT OF UP TO
6 YEARS



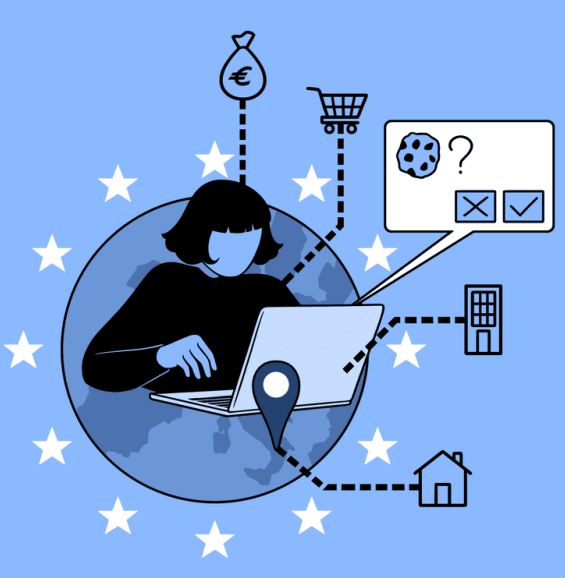
HEFTY FINE OF
PHP500,000

Global Data Privacy Regulations

Many countries now enforce strict data protection laws.

Major Regulations:

- GDPR – European Union data protection law
- CCPA – California consumer privacy law
- Data Privacy Act – Philippines



General Data Protection Regulation (GDPR)

*[ˈjɛn-rəl ˈdā-tə prə-ˈtɛk-shən
ˌre-gyə-ˈlā-shən]*

Guidelines for the collection and processing of personal data of individuals within the European Union.



Data Breaches

A data breach occurs when confidential information is exposed.

Data Breach:

- Unauthorized access to sensitive data
- Often caused by cyber attacks or misconfiguration



Introduction to Blockchain

Blockchain is a distributed system for recording transactions securely.

Blockchain:

- Decentralized digital ledger
- Records transactions across multiple nodes
- Ensures transparency and immutability



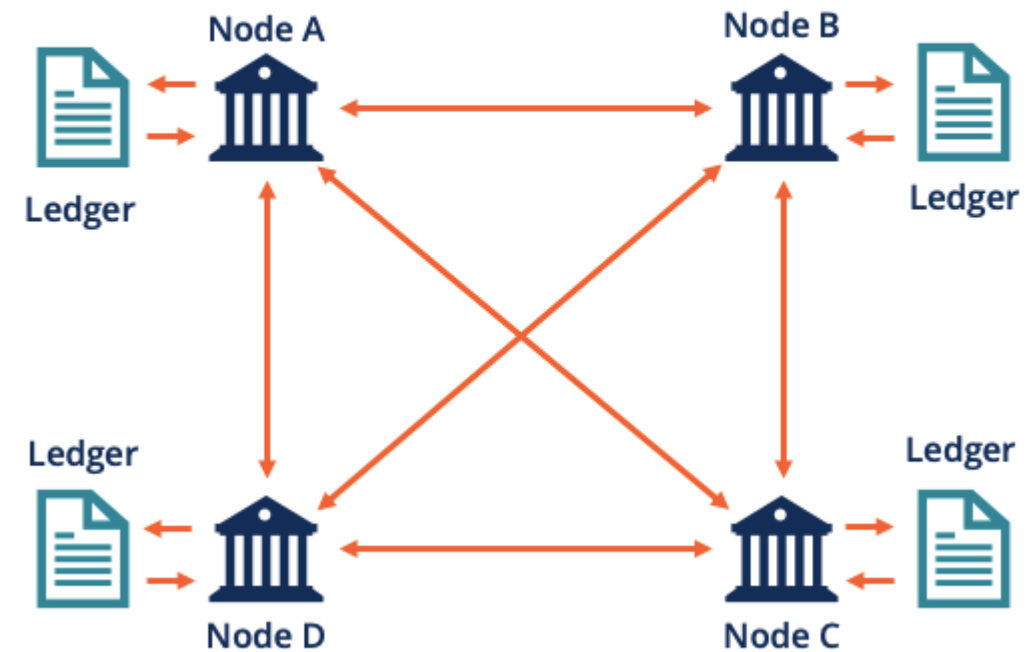
How Blockchain Works?

Blockchain records transactions in blocks linked together.

Key Components:

- Blocks – records of transactions
- Nodes – computers maintaining the network
- Consensus – agreement on valid transactions

Distributed Ledgers



Smart Contracts

Smart contracts automate agreements using blockchain.

Smart Contracts:

- Self-executing digital contracts
- Trigger actions when conditions are met



Web3 Concept

Web3 represents a decentralized vision of the internet.

Web3:

- Decentralized applications
- Blockchain-based ownership
- User-controlled data



Benefits of Blockchain

Blockchain offers several advantages in digital systems.

Advantages:

- Transparency
- Security
- Decentralization



Blockchain Limitations

Despite its advantages, blockchain faces major challenges.

Limitations:

- High energy consumption
- Scalability issues
- Regulatory uncertainty



Blockchain Security Risks

While blockchain is designed to be secure, it is not immune to attacks. Vulnerabilities often arise from poor implementation rather than the technology itself.

Blockchain Security Risks:

- Smart contract vulnerabilities – coding flaws that attackers exploit
- 51% attack – when a group controls majority of network power
- Private key theft – loss of wallet credentials
- Exchange hacking – attacks targeting crypto platforms

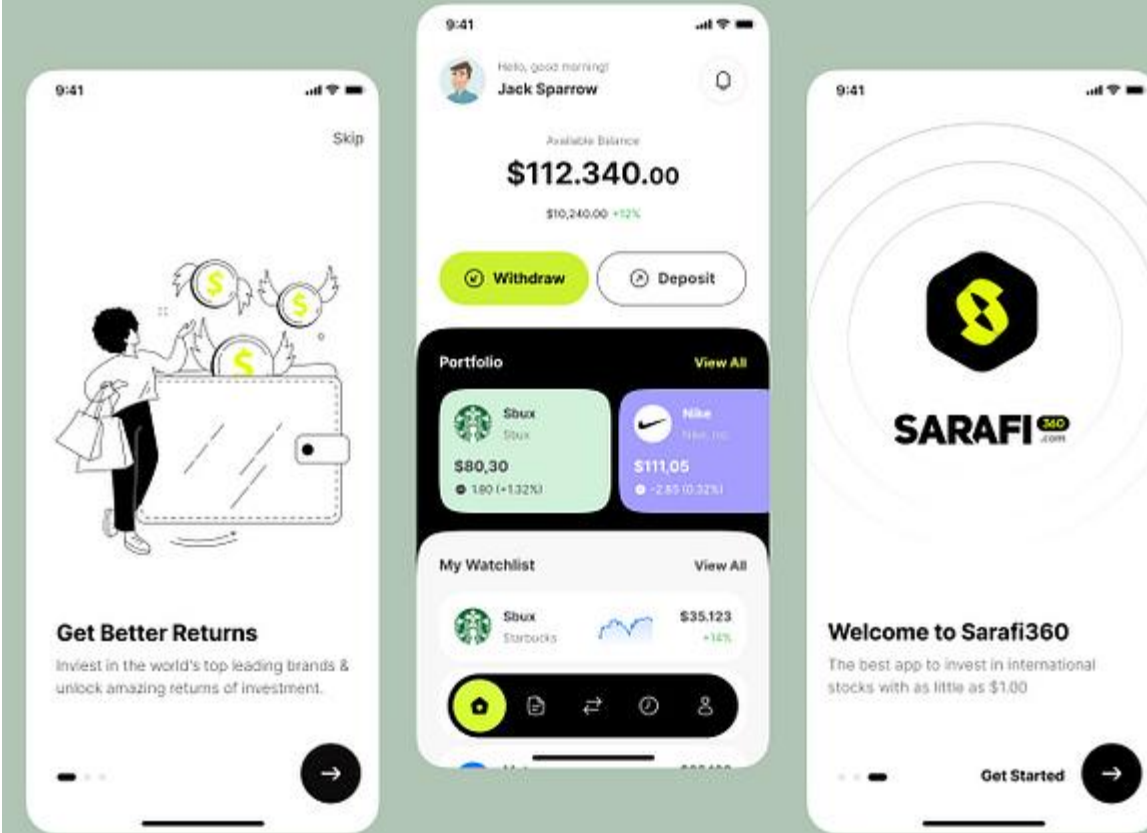


Cryptocurrency

Cryptocurrency is one of the most well-known applications of blockchain technology.

Cryptocurrency:

- Digital currency using cryptography
- Transactions verified through blockchain networks
- Examples: Bitcoin, Ethereum



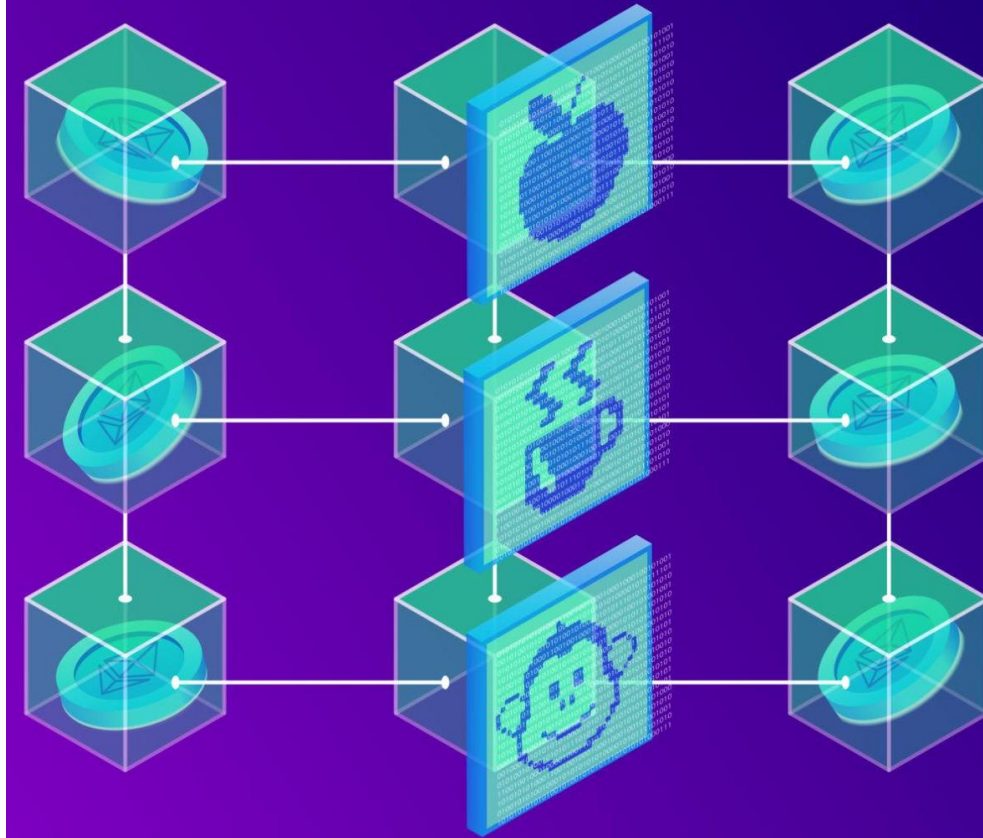
Non-Fungible Tokens (NFTs)

NFTs represent unique digital ownership recorded on a blockchain.

NFT:

- Unique digital asset
- Ownership verified through blockchain
- Often used for digital art and collectibles

NFT : Non-Fungible Token



Decentralized Finance (DeFi)

Decentralized Finance uses blockchain to replicate traditional financial services.

DeFi:

- Financial services without banks
- Uses smart contracts for transactions
- Includes lending, trading, and insurance

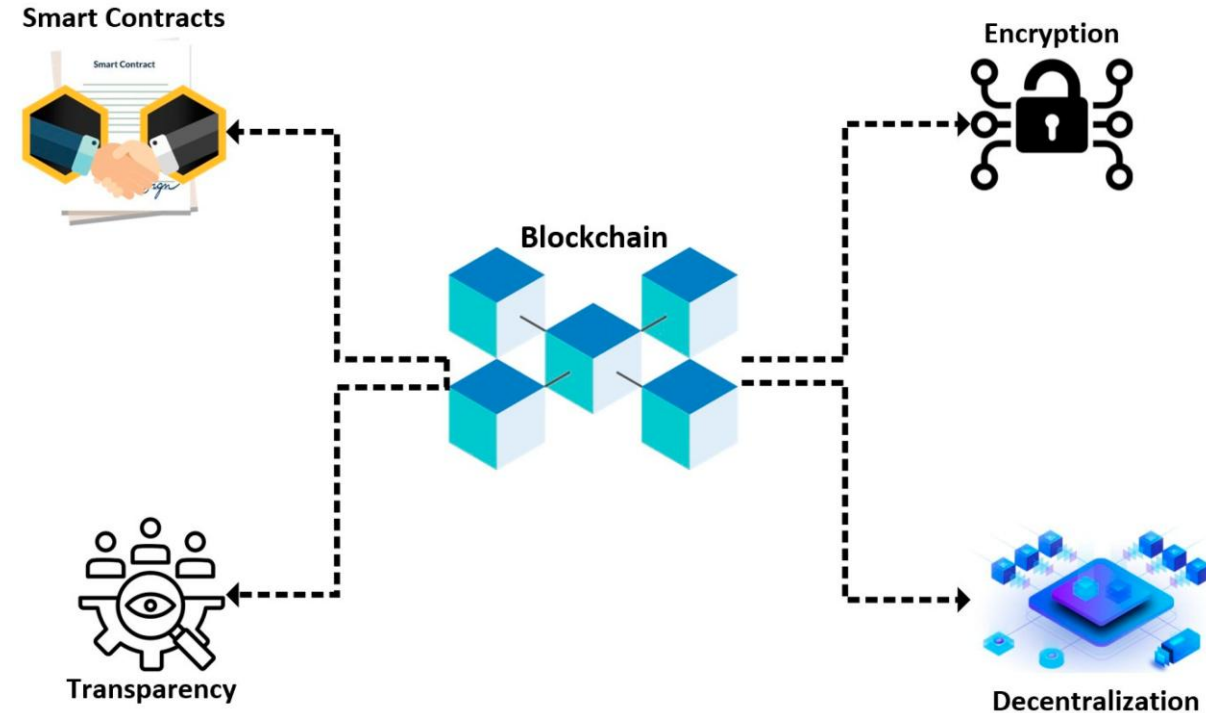


Governance Challenges in Blockchain

Decentralized systems introduce new governance questions.

Governance Challenges:

- Lack of central authority
- Difficulty enforcing regulations
- Conflicts among network participants



Ethical Concerns in Web3

Emerging technologies always raise ethical and social questions.

Ethical Issues:

- Financial speculation risks
- Environmental impact of mining
- Fraud and scams

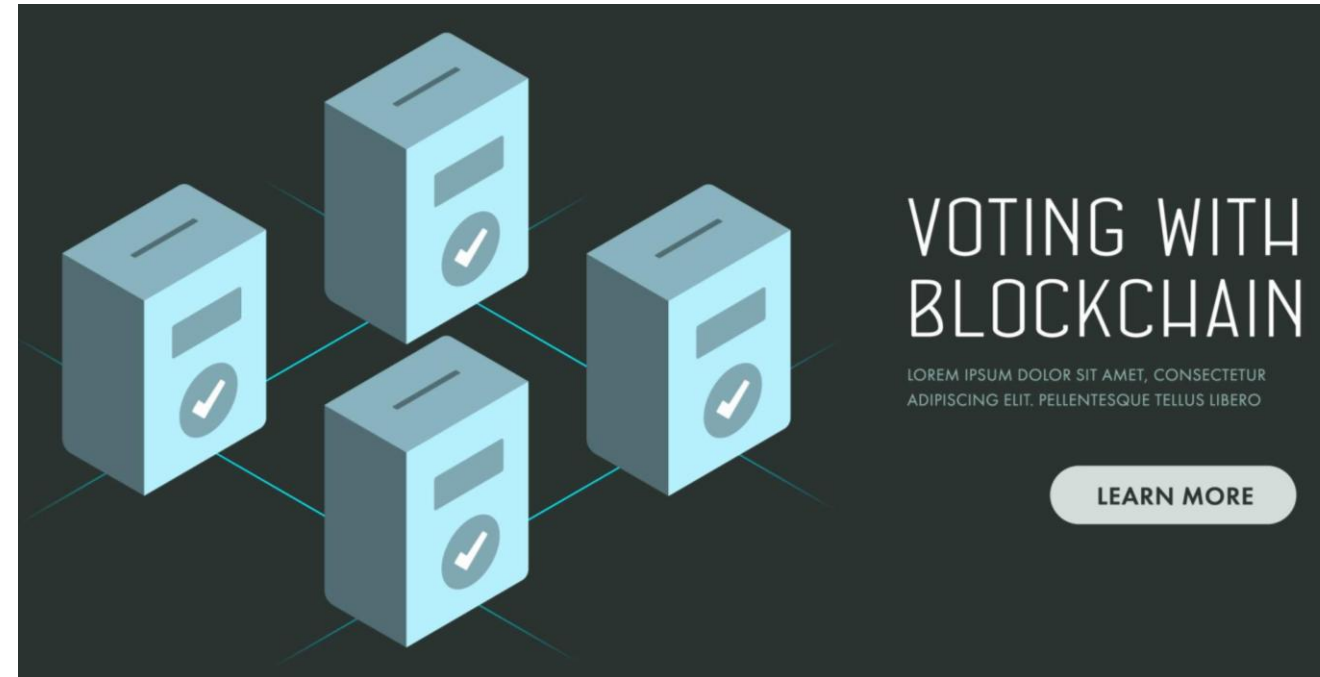


Future of Blockchain and Web3

Blockchain technologies continue evolving across industries.

Emerging Applications:

- Supply chain transparency
- Digital identity verification
- Secure voting systems



Framework for Evaluating Emerging Technologies

When analyzing new technologies, we must evaluate multiple factors.

Evaluation Framework:

- Technical feasibility
- Business value
- Security and privacy risks
- Regulatory implications

	Catastrophic 5	Major 4	Moderate 3	Minor 2	Negligible 1
Is practically impossible and has never occurred	25	20	15	10	5
Has not been known to occur after many years	15	12	9	6	3
Might be occur at sometimes in future	10	8	6	4	2
Has a good chance of occurring and is not unusual	5	4	3	2	1
LIKELIHOOD Description	Most Likely	Possible	Conceivable	Remote	Inconceivable



Final Synthesis

Data governance and blockchain illustrate how technology, law, and ethics intersect.

Understanding these issues prepares IT professionals to design responsible digital systems.

Key Takeaways:

- Data governance ensures responsible data management
- Privacy laws protect user information
- Blockchain enables decentralized systems Web3 introduces new opportunities and risks
- Ethical thinking is essential in emerging technologies



Module 4 Activity (Individual)

Data Governance Compliance Review

Guidelines/Requirements:

1. Research for data governance policy of any company (Facebook / Meta, Google, Amazon, Gcash, BPI, Any Government websites, Any company with available privacy policy)
2. Analyze an its data governance policy (Privacy policy, Data governance practices, Data collection and usage policies)
3. Write a short analysis (<500 words) covering:

A. Organization Overview

Brief background of the company
What type of data they collect

B. Data Governance Practices

How they store and manage data
Access control and security measures

C. Strengths

What the company does well in protecting data

D. Weaknesses or Risks

Possible privacy risks or governance gaps

E. Recommendations

2–3 improvements for better data governance

Criteria	Description	Points
Understanding of Data Governance	Shows clear understanding of how data governance works	5
Analysis of Organization	Correctly explains company data practices	5
Identification of Strengths & Weaknesses	Clearly identifies governance strengths and risks	5
Recommendations	Practical and realistic improvement suggestions	5
Clarity and Organization	Clear writing, logical structure, proper sources	5

